
Computer Vision

Chapter 7 (part 2): Object detection

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Contents

- Window-based generic object detection: basic pipeline
 - Boosting classifiers
- Face detection as case study
 - Haar features
 - Adaboost
- SVM + HOG for human detection as case study
- Object proposals
- [DPM]
- Evaluation



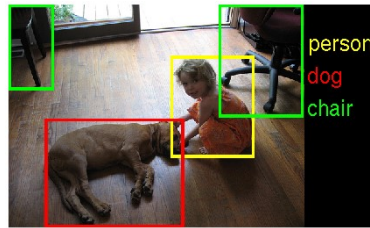
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Object Detection

- **Problem:** Detecting and localizing generic objects from various categories, such as cars, people, etc.

- **Challenges:**

- Illumination,
- viewpoint,
- deformations,
- Intra-class variability



Window-based generic object detection

Basic pipeline

Basic Pipeline

- Build/train object model
 - Choose a representation
 - Learn or fit parameters of model / classifier
- Generate candidates in new image
- Score the candidates



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Basic Pipeline

- **Build/train object model**
 - Choose a representation
 - Learn or fit parameters of model / classifier

Given the representation,
train a binary classifier



Car/non-car
Classifier



No/Yes to car.

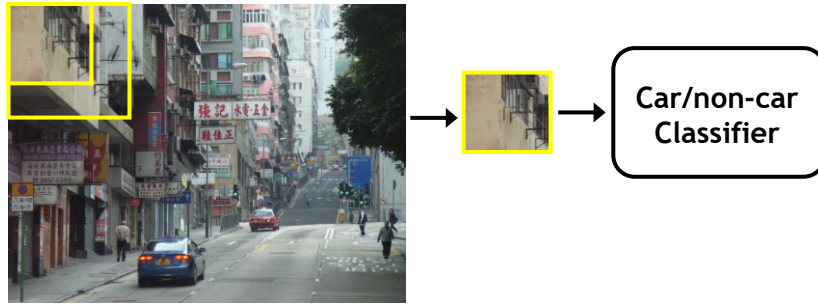


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Basic Pipeline

- **Generate candidates** in new image
- **Score** the candidates



Window-based models: Window size?

- Slide through the image and check if there is an object at every location



YES!! Person match found

Window-based models: Window size ?

- But what if we were looking for buses?

No bus found!



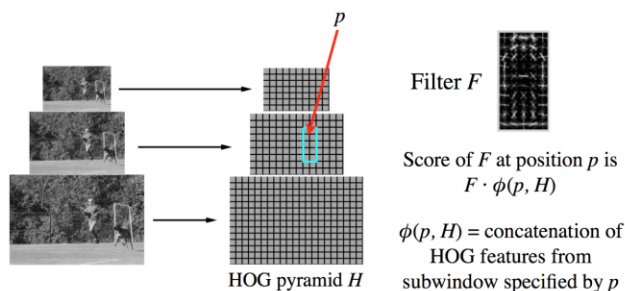
- We will never find the object if we don't choose our window size wisely!

Bus found



Multi-scale sliding window

- Work with multiple window sizes
- Create a feature pyramid



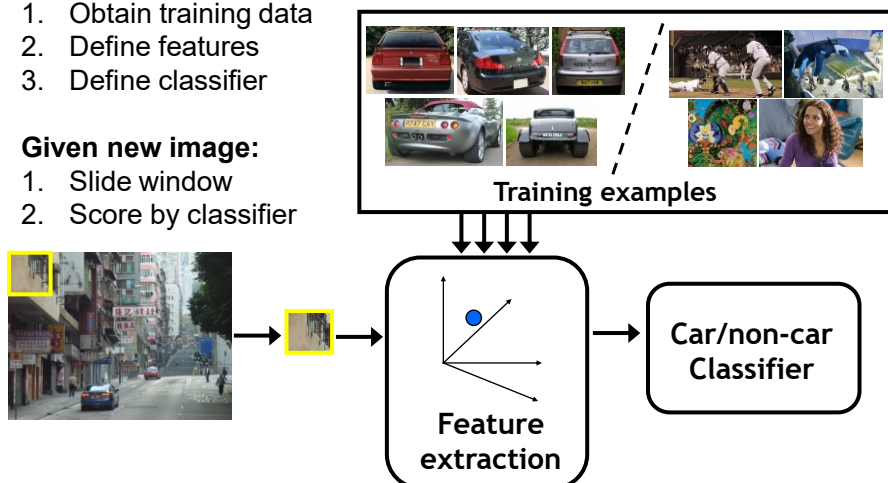
Window-based object detection: recap

Training:

1. Obtain training data
2. Define features
3. Define classifier

Given new image:

1. Slide window
2. Score by classifier



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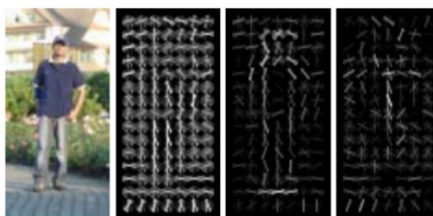
Slide: Kristen Grauman

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Features

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- HOG



- Bags of visual words

Bag of 'words'



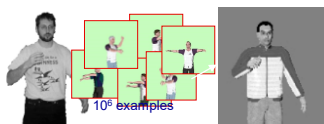
- Haar features, ...



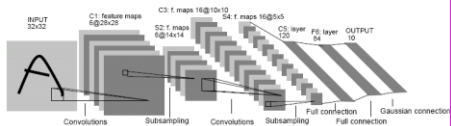
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Discriminative classifier construction

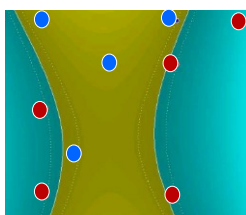
Nearest neighbor



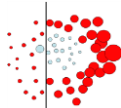
Neural networks



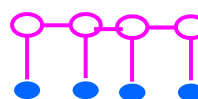
Support Vector Machines



Boosting

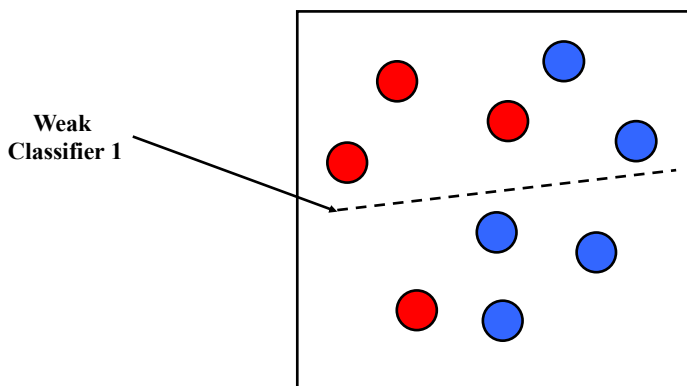


Conditional Random Fields

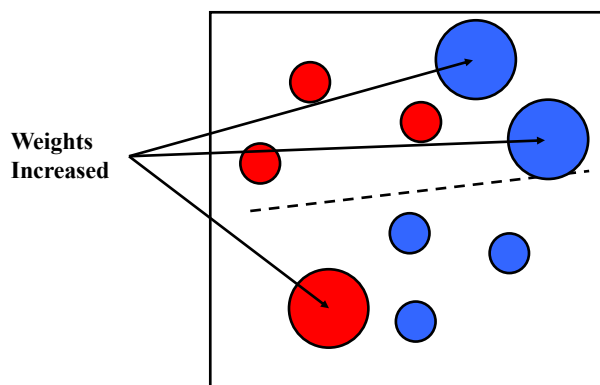


Boosting classifiers

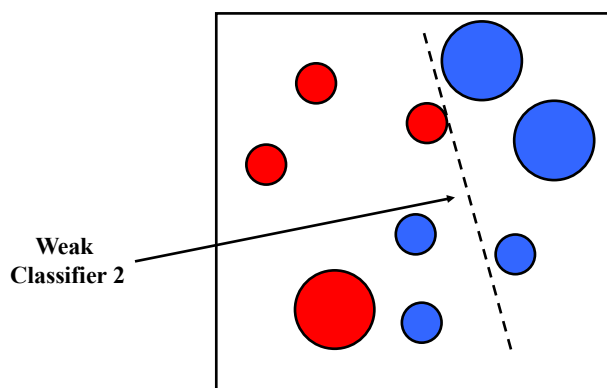
Boosting intuition



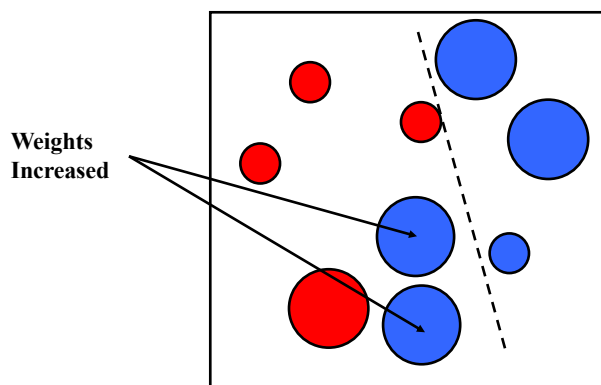
Boosting illustration



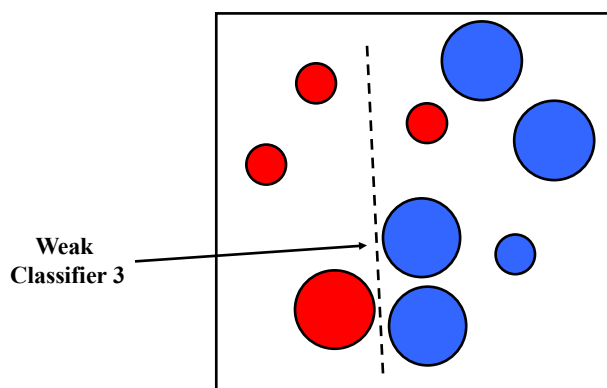
Boosting illustration



Boosting illustration

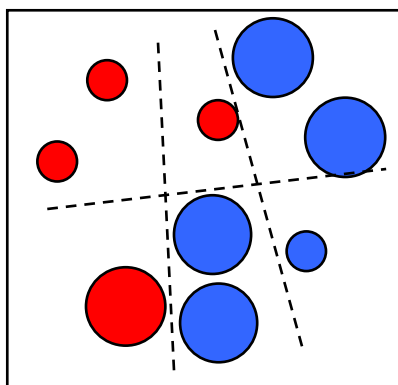


Boosting illustration



Boosting illustration

**Final classifier is
a combination of weak
classifiers**



Boosting: training

- Initially, weight each training example equally
- In each boosting round:
 - Find the **weak learner** that achieves the lowest *weighted* training error
 - **Raise weights of training examples misclassified** by current weak learner
- Compute final classifier as linear combination of all weak learners
 - (weight of each learner is directly proportional to its accuracy)
- Exact formulas for re-weighting and combining weak learners **depend on the particular boosting scheme** (e.g., *AdaBoost*)



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Face detection as case study



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Viola-Jones face detector

ACCEPTED CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION 2001

Rapid Object Detection using a Boosted Cascade of Simple Features

Paul Viola
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Mitsubishi Electric Research Labs
201 Broadway, 8th FL
Cambridge, MA 02139

Michael Jones
mjones@crl.dec.com
Compaq CRL
One Cambridge Center
Cambridge, MA 02142

Abstract

This paper describes a machine learning approach for vi-

tested at 15 frames per second on a conventional 700 MHz Intel Pentium III. In other face detection systems, auxiliary information, such as image differences in video sequences,



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Viola-Jones face detector

Main idea:

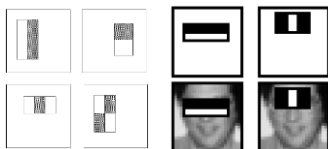
- Represent **local texture with efficiently computable “rectangular” features** within window of interest
- Select **discriminative features** to be weak classifiers
- Use **boosted combination** of them as final classifier
- Form a **cascade of such classifiers**, rejecting clear negatives quickly



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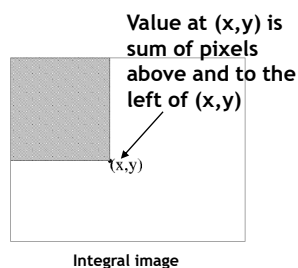
Viola-Jones detector: features



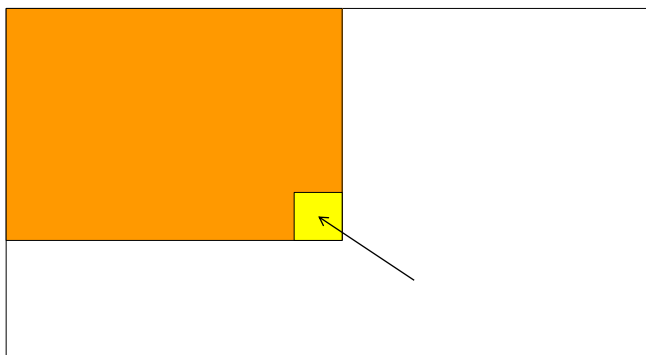
- “Rectangular” filters

Feature output is difference between adjacent regions

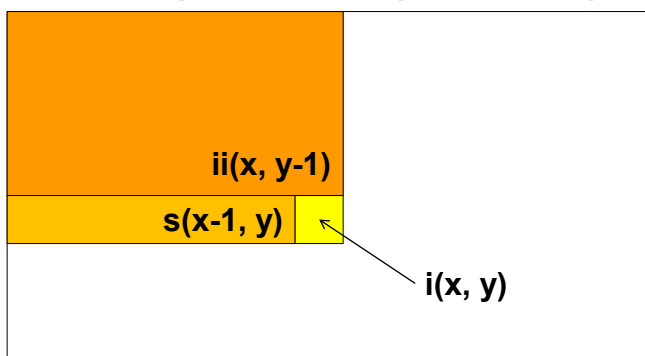
- Efficiently computable with **integral image**: any sum can be computed in constant time.



Computing the integral image



Computing the integral image

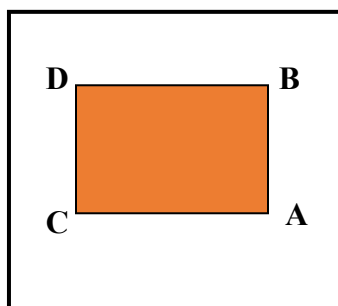


- Cumulative row sum: $s(x, y) = s(x-1, y) + i(x, y)$
- Integral image: $ii(x, y) = ii(x, y-1) + s(x, y)$

Computing sum within a rectangle

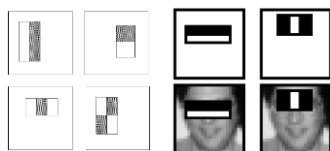
- Let **A,B,C,D** be the values of the integral image at the corners of a rectangle
- Then the sum of original image values within the rectangle can be computed as:

$$\text{sum} = A - B - C + D$$



- Only 3 additions are required for any size of rectangle!

Viola-Jones detector: features

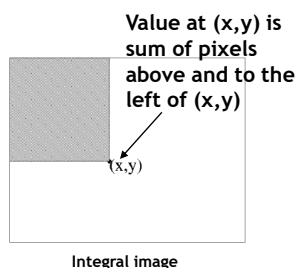


- “Rectangular” filters

Feature output is difference between adjacent regions

- Efficiently computable with integral image: any sum can be computed in constant time

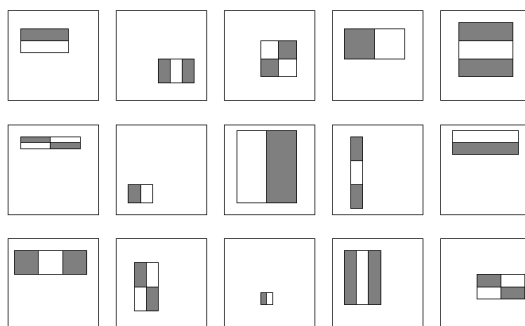
Avoid scaling images → scale features directly for same cost



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Viola-Jones detector: features



Considering all possible filter parameters: position, scale, and type:

180,000+ possible features associated with each 24 x 24 window

Which subset of these features should we use to determine if a window has a face?

Use AdaBoost both to select the informative features and to form the classifier

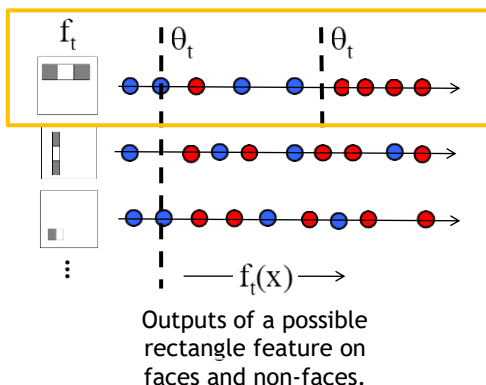


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Viola-Jones detector: AdaBoost

- Want to select the single rectangle feature and threshold that best separates **positive** (faces) and **negative** (non-faces) training examples, in terms of *weighted error*.



Resulting weak classifier:

$$h_t(x) = \begin{cases} +1 & \text{if } f_t(x) > \theta_t \\ -1 & \text{otherwise} \end{cases}$$

For next round, reweight the examples according to errors, choose another filter/threshold combo.



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- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \dots, T$:

1. Normalize the weights,

$$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$$

so that w_t is a probability distribution.

2. For each feature, j , train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t , $\epsilon_j = \sum_i w_i |h_j(x_i) - y_i|$.
3. Choose the classifier, h_t , with the lowest error ϵ_t .
4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$

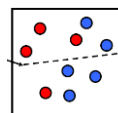
where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.

- The final strong classifier is:

$$h(x) = \begin{cases} 1 & \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

Start with uniform weights on training examples



For T rounds

$\{x_1, \dots, x_n\}$

Evaluate *weighted error* for each feature, pick best.

Re-weight the examples:

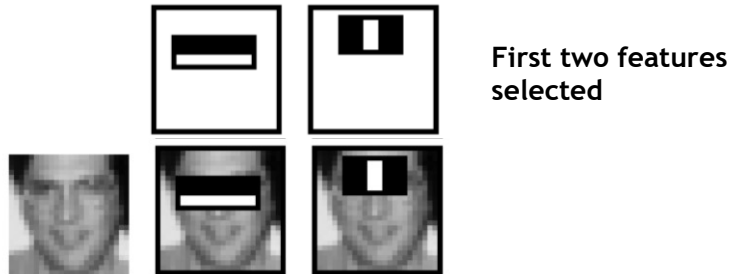
- Incorrectly classified -> more weight
- Correctly classified -> less weight

Final classifier is combination of the weak ones, weighted according to error they had.

CHNOLOGY

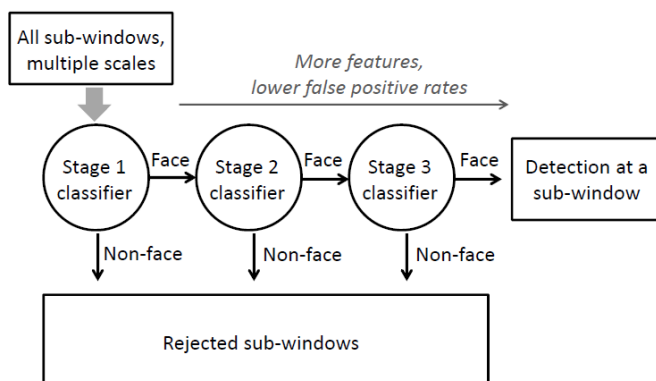
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Viola-Jones Face Detector: Results



-
- Even if the filters are fast to compute, each new image has a lot of possible windows to search.
 - How to make the detection more efficient?

Cascading classifiers for detection



- Form a *cascade* with **low false negative rates** early on
- Apply less accurate but faster classifiers first **to immediately discard** windows that **clearly appear to be negative**



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Training the cascade

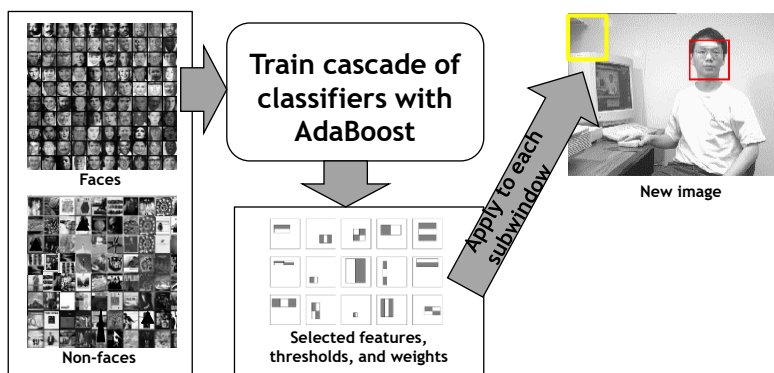
- Set **target detection** and **false positive rates** for each stage
- Keep **adding features** to the current stage until its target rates have been met
 - Need to **lower AdaBoost threshold to maximize detection** (as opposed to minimizing total classification error)
 - Test on a *validation set*
- If the **overall false positive rate is not low enough**, then **add another stage**
- Use **false positives** from current stage as **the negative training examples for the next stage**



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Viola-Jones detector: summary



- Train with 5K positives, 350M negatives
- Real-time detector using 38 layer cascade
- 6061 features in all layers



[Implementation available in OpenCV]
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Viola-Jones detector: summary

- A seminal approach to real-time object detection
 - 16,165 citations and counting
- Training is slow, but detection is very fast
- Key ideas
 - Integral images for fast feature evaluation
 - Boosting for feature selection
 - Attentional cascade of classifiers for fast rejection of non-face windows

P. Viola and M. Jones. [Rapid object detection using a boosted cascade of simple features](#). CVPR 2001.

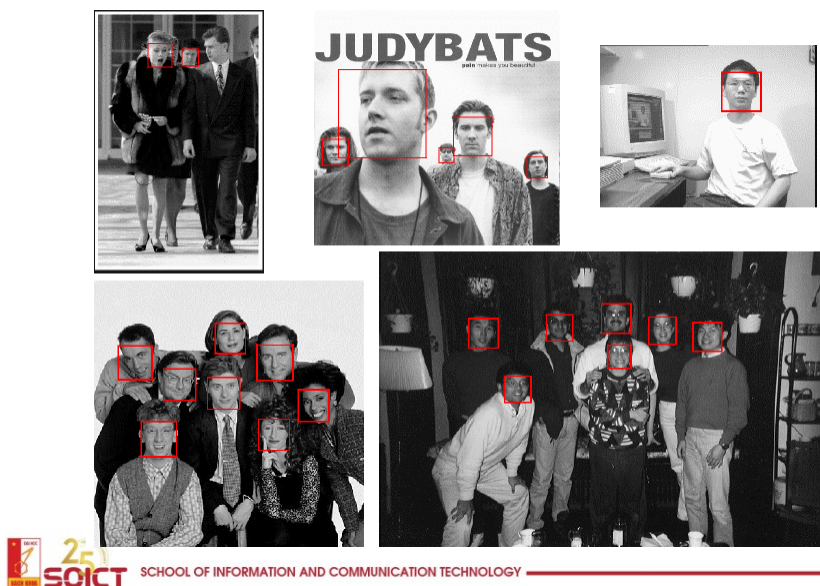
P. Viola and M. Jones. [Robust real-time face detection](#). IJCV 57(2), 2004.



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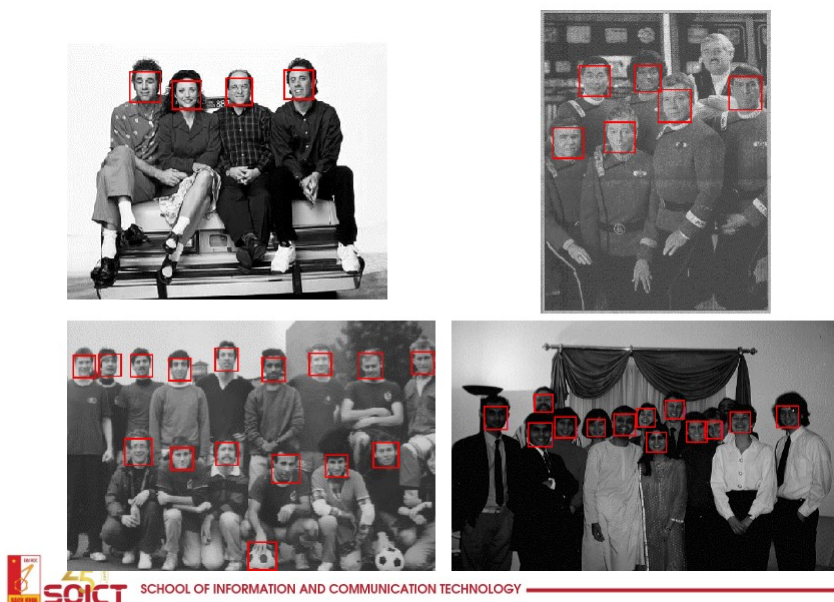
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Viola-Jones Face Detector: Results



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Viola-Jones Face Detector: Results



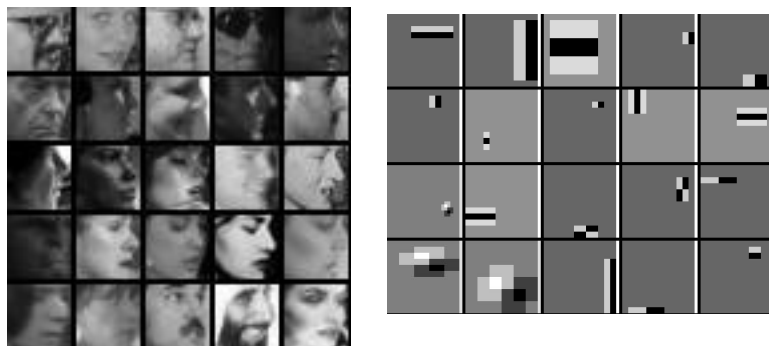
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Viola-Jones Face Detector: Results

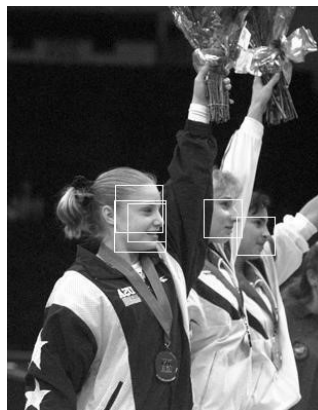


Detecting profile faces?

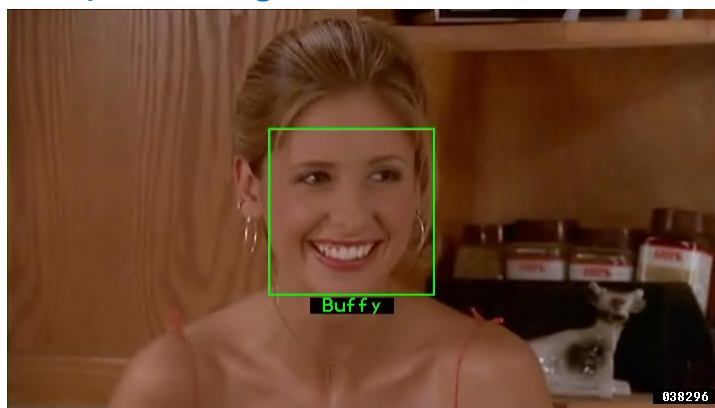
Can we use the same detector?



Viola-Jones Face Detector: Results



Example using Viola-Jones detector



Frontal faces detected and then tracked, character names inferred with alignment of script and subtitles.

Everingham, M., Sivic, J. and Zisserman, A.

"Hello! My name is... Buffy" - Automatic naming of characters in TV video, BMVC 2006. <http://www.robots.ox.ac.uk/~vgg/research/nface/index.html>



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Google now erases faces, license plates on Map Street View

By Elinor Mills, CNET News.com
Friday, August 24, 2007 01:37 PM

Google has gotten a lot of flack from privacy advocates for photographing faces and license plate numbers and displaying them on the Street View in Google Maps. Originally, the company said only people who identified themselves could ask the company to remove their image.

But Google has quietly changed that policy, partly in response to criticism, and now anyone can alert the company and have an image of a license plate or a recognizable face removed, not just the owner of the face or car, says Marissa Mayer, vice president of search products and user experience at Google.

"It's a good policy for users and also clarifies the intent of the product," she said in an interview following her keynote at the Search Engine Strategies conference in San Jose, Calif., Wednesday.

The policy change was made about 10 days after the launch of the product in late May, but was not publicly announced, according to Mayer. The company is removing images only when someone notifies them and not proactively, she said. "It was definitely a big policy change inside."

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- » Google begins search for Middle East lobbyist
- » Google still thinks it can change China

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Technology

Google street view blurs face of cow to protect its identity



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Consumer application: iPhoto

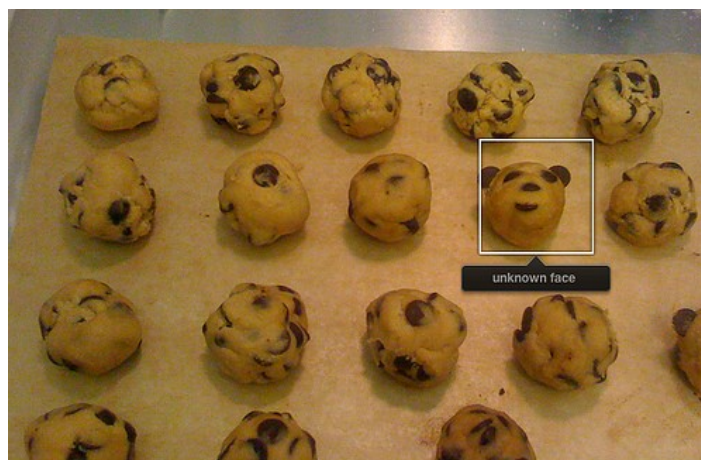


<http://www.apple.com/ilife/iphoto/>
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Slide credit: Lana Lazebnik

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Consumer application: iPhoto



Things iPhoto thinks are faces



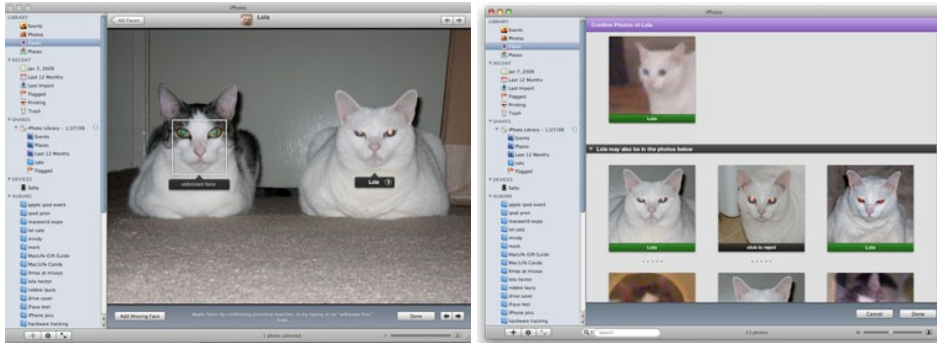
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Slide credit: Lana Lazebnik

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Consumer application: iPhoto

- Can be trained to recognize pets!



http://www.maclife.com/article/news/iphotos_faces_recognizes_cats



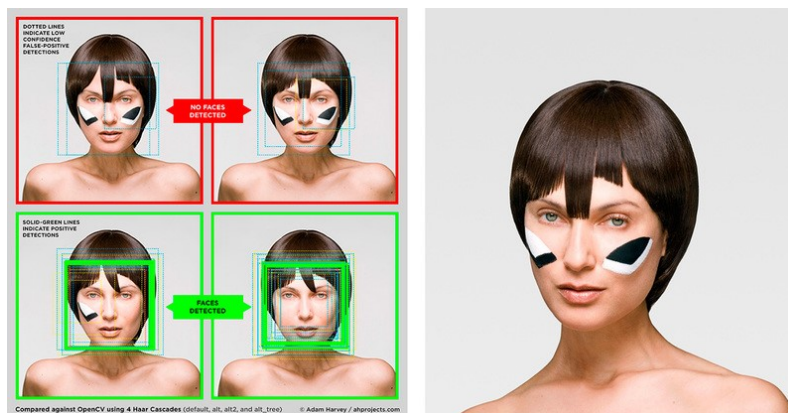
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Slide credit: Lana Lazebnik

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Privacy Gift Shop – CV Dazzle

- <http://www.wired.com/2015/06/facebook-can-recognize-even-dont-show-face/>
- Wired, June 15, 2015



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Boosting: pros and cons

- Advantages of boosting
 - Integrates classification with feature selection
 - Complexity of training is linear in the number of training examples
 - Flexibility in the choice of weak learners, boosting scheme
 - Testing is fast
 - Easy to implement
- Disadvantages
 - Needs many training examples
 - Other discriminative models may outperform in practice (SVMs, CNNs,...)
 - especially for many-class problems

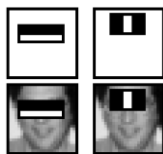


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Slide credit: Lana Lazebnik

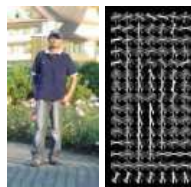
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Window-based models: Two case studies



Boosting + face
detection

Viola & Jones



SVM + person
detection

e.g., Dalal & Triggs



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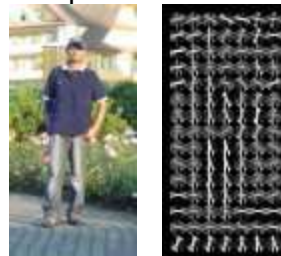
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SVM + HOG for human detection as case study



Person detection with HoG's & linear SVM's

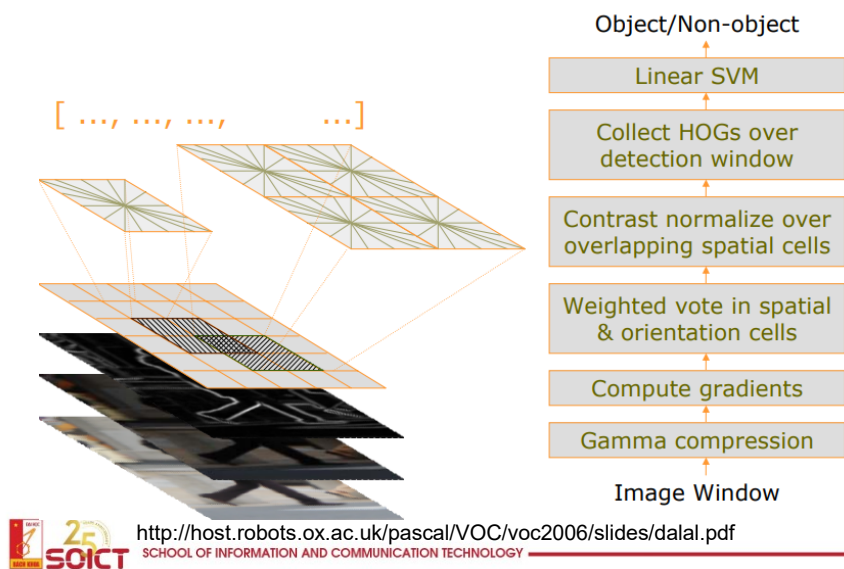
- Histogram of oriented gradients (HoG):
 - Map each grid cell in the input window to a histogram counting the gradients per orientation.
- Train a linear SVM
 - using training set of pedestrian vs. non-pedestrian windows.



<https://lear.inrialpes.fr/people/triggs/pubs/Dalal-cvpr05.pdf>



Person detection with HoG's & linear SVM's



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Person detection with HoGs & linear SVMs



- For more detail about HoG:
 - Histograms of Oriented Gradients for Human Detection, [Navneet Dalal](#), [Bill Triggs](#), International Conference on Computer Vision & Pattern Recognition - June 2005
 - <http://lear.inrialpes.fr/pubs/2005/DT05/>

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Key Descriptor Parameters

Class	Window Size	Avg. Size	# of Orientation Bins	Orientation Range	Gamma Compression	Normalisation Method
Person	64×128	Height 96	9	0°–180°	√RGB	L2-Hys
Car	104×56	Height 48	18	0°–360°	√RGB	L1-Sqrt
Bus	120×80	Height 64	18	0°–360°	√RGB	L1-Sqrt
Motorbike	120×80	Width 112	18	0°–360°	√RGB	L1-Sqrt
Bicycle	104×64	Width 96	18	0°–360°	√RGB	L2-Hys
Cow	128×80	Width 56	18	0°–360°	√RGB	L2-Hys
Sheep	104×60	Height 56	18	0°–360°	√RGB	L2-Hys
Horse	128×80	Width 96	9	0°–180°	RGB	L1-Sqrt
Cat	96×56	Height 56	9	0°–180°	RGB	L1-Sqrt
Dog	96×56	Height 56	9	0°–180°	RGB	L1-Sqrt

Window-based detection: strengths

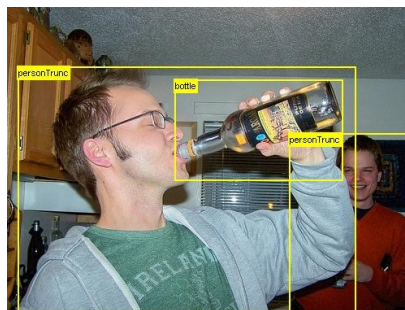
- Sliding window detection and global appearance descriptors:
 - Simple detection protocol to implement
 - Good feature choices critical
 - Past successes for certain classes

Window-based detection: Limitations

- High computational complexity
 - For example: 250,000 locations x 30 orientations x 4 scales = 30,000,000 evaluations!
 - If training binary detectors independently, means cost increases linearly with number of classes
- With so many windows, false positive rate better be low

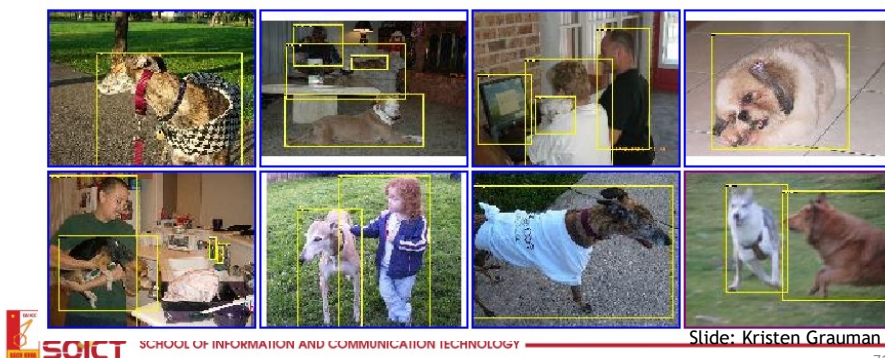
Limitations (continued)

- Not all objects are “box” shaped

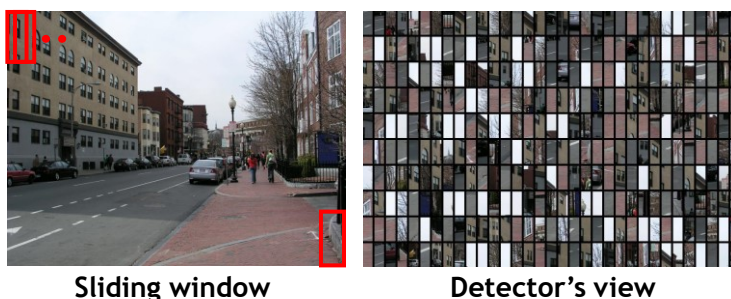


Limitations (continued)

- **Non-rigid, deformable** objects **not captured well** with representations assuming a fixed 2d structure; or must assume fixed viewpoint
- **Objects with less-regular textures** not captured well with holistic appearance-based descriptions



Limitations (continued)



If considering windows in isolation,
context is lost

Limitations (continued)

- In practice, often entails large, cropped training set (**expensive**)
- Requiring good match to a global appearance description can lead to **sensitivity to partial occlusions**

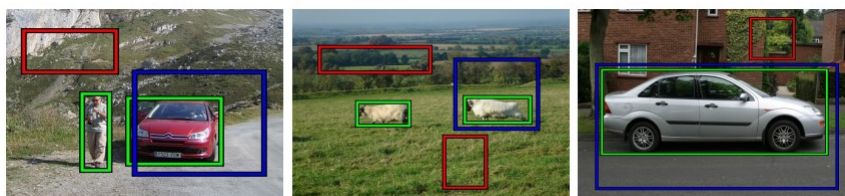


Object proposals

Object proposals

Main idea:

- Learn to generate category-independent regions/boxes that have **object-like** properties.
- Let object detector **search over “proposals”**, not exhaustive sliding windows

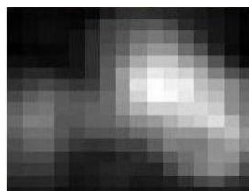


Alexe et al. Measuring the objectness of image windows, PAMI 2012

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Object proposals



Multi-scale
saliency



Color
contrast



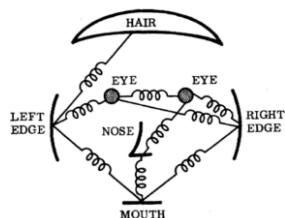
Alexe et al. Measuring the objectness of image windows, PAMI 2012

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Deformable Part Model (DPM)

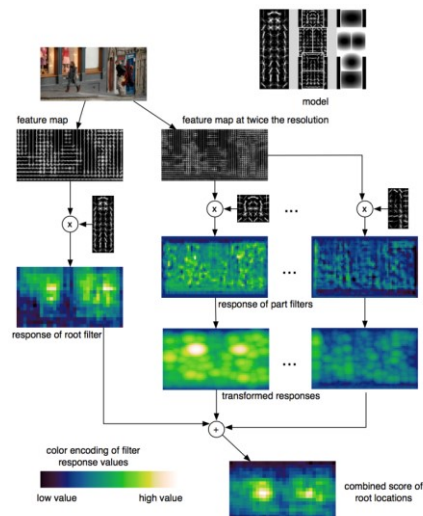
- Represents an object as a **collection of parts** arranged in a deformable configuration
- Each part represents **local appearances**
- Spring-like connections between certain pairs of parts



Fischler and Elschlager, Pictorial Structures, 1973



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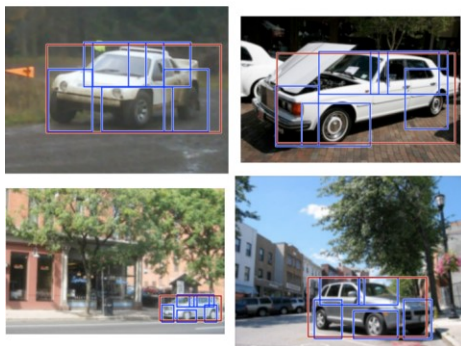


Felzenszwalb et al., PAMI 2010

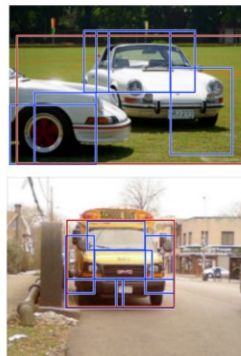
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Deformable Part Model (DPM)

high scoring true positives



high scoring false positives



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Deformable Part Model (DPM)

- References

- Pedro F. Felzenszwalb & Daniel P. Huttenlocher, Pictorial Structures for Object Recognition, IJCV 2005
 - <https://www.cs.cornell.edu/~dph/papers/pict-struct-ijcv.pdf>
- P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan. Object detection with discriminatively trained part based models. IEEE Transactions on Pattern Analysis and Machine Intelligence, 32(9):1627–1645, 2010



Object detection: Evaluation



Object Detection Benchmarks

- PASCAL VOC Challenge
- ImageNet Large Scale Visual Recognition Challenge (ILSVR)
 - 200 Categories for detection
- Common Objects in Context (COCO)
 - 80 Object categories



How do we evaluate object detection?



- predictions
- ground truth

True positive:

- The overlap of the prediction with the ground truth is **MORE** than a threshold value (0.5)

How do we evaluate object detection?



- predictions
- ground truth

True positive:

False positive:

- The overlap of the prediction with the ground truth is **LESS** than a threshold value (0.5)

How do we evaluate object detection?



- predictions
- ground truth

True positive:

False positive:

False negative:

- The objects that our model doesn't find

How do we evaluate object detection?



— predictions
— ground truth

True positive:

False positive:

False negative:

- The objects that our model doesn't find

What is a **True Negative**?

	Predicted 1	Predicted 0
True 1	true positive	false negative
True 0	false positive	true negative

	Predicted 1	Predicted 0
True 1	TP	FN
True 0	FP	TN

	Predicted 1	Predicted 0
True 1	hits	misses
True 0	false alarms	correct rejections

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

How do we evaluate object detection?



— predictions
— ground truth

True positive: 1
False positive: 2
False negative: 1

So what is the
- precision?
- recall?

Precision versus recall

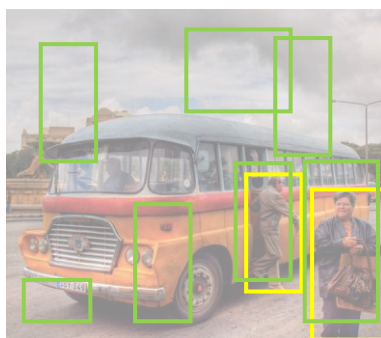
- Precision:
 - how many of the object detections are correct?

$$precision = \frac{TP}{TP + FP}$$

- Recall:
 - how many of the ground truth objects can the model detect?
 - True Positive Rate (TPR)

$$recall = \frac{TP}{TP + FN}$$

- In reality, our model makes a lot of predictions with varying scores between 0 and 1



— predictions
— ground truth

Here are all the boxes that are predicted with **score > 0**.

This means that our

- **Recall is perfect!**
- But our **precision is BAD!**

How do we evaluate object detection?

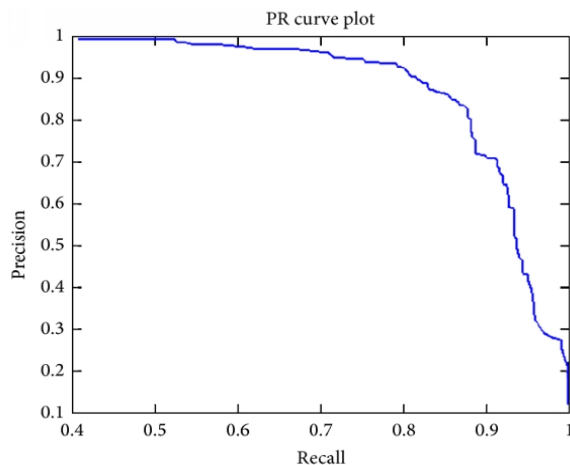


— predictions
— ground truth

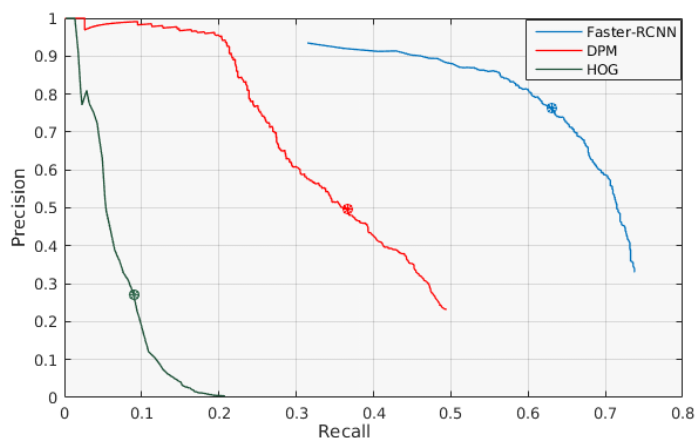
Here are all the boxes that are predicted with **score > 0.5**

We are setting a **threshold** of 0.5

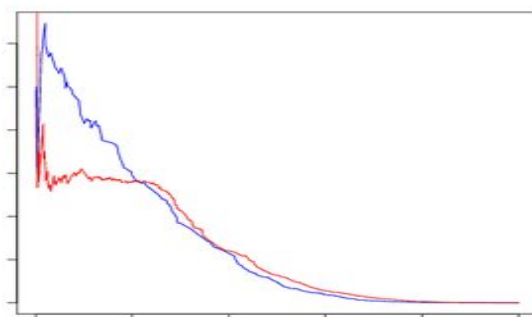
Precision – recall curve (PR curve)



Which model is the best?

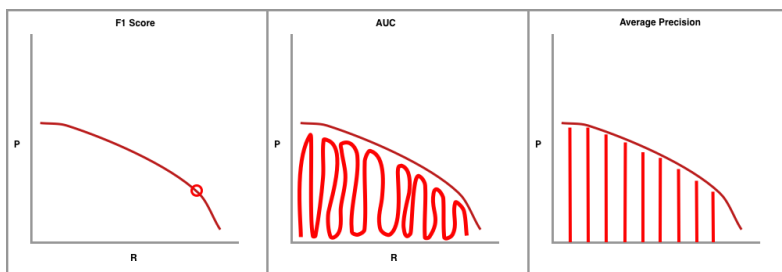


Which model is the best?



- **Area under curve (AUC), average precision (AP)**
- **F1-score** (highest value at optimal confidential score)

Which model is the best?



AP: The metric calculates the average precision (AP) for each class individually across all of the IoU thresholds

$$AP = \frac{1}{11} \sum_{r \in \{0.0, 0.1, 0.2, \dots, 0.9, 1\}} p_{inter-p}(r)$$

mAP: the average of AP

$$= \frac{1}{11} (1 + 1 + 1 + 1 + 0.67 + 0.67 + 0.67 + 0.5 + 0.5 + 0.5 + 0.5) \\ \approx 0.728$$

Summary

- Object detection as classification task
 - Boosting (face detection ex)
 - Support vector machines and HOG (human detection ex)
 - Sliding window search paradigm
 - Pros and cons
 - Speed up with attentional cascade
 - Object proposals, proposal regions as alternative



References

Most of these slides were adapted from:

1. Kristen Grauman (CS 376: Computer Vision, Spring 2018, The University of Texas at Austin)

